

Customer Churn Prediction Using Machine Learning Models

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Abstract—Customer churn is a critical challenge for telecommunication companies, impacting profitability and growth. This project presents an analysis using machine learning models such as Logistic Regression, Decision Tree, Random Forest, and XGBoost to predict customer churn. The study highlights key influential features and evaluates model performance based on precision, recall, f1-score of class 1 and accuracy, addressing challenges like class imbalance.

Index Terms—Customer Churn, Telco Company, Machine Learning, Logistic Regression, Decision Tree, Random Forest, XGBoost, Class Imbalance.

I. INTRODUCTION

Customer churn refers to the phenomenon that customers stop using a company's products or services during a certain period of time. While churn is a natural part of any business cycle, high customer churn rates can be a significant problem, indicating poor customer retention and low customer loyalty. A high churn rate often relates to various underlying factors, such as poor product or service quality, bad product-market fit, price disadvantage, limited customer support, intense competition, etc. Customer churn is a particularly pressing challenge in telecommunications companies, which are service providers in the networking industry that facilitate the electronic transmission of information through telephony and data communication services. With the advent of mobile telephony, most telcos also provide internet service. Telco customer churn rates are among the highest in any industry, averaging 31 percent in 2024 (Tessitore). This is due to the inherent nature of this sector, such as subscription-based plan, highly competitive market of providers offering similar products, rapid technological advancement, etc. Customer churn is not only loss of customers, but also affects the long-term profitability and sustainability of a company. Thus, understanding and predicting customer churn is very important, because it allows companies to proactively address at-risk customers.

The primary objective of this project is to build predictive machine learning models that accurately classify customers into "churn" and "non-churn" categories. This project will also analyze influential factors contributing to customer churn, and discuss the relationships between these factors and churn. The insights derived can help support data-driven strategies to reduce customer churn rates.

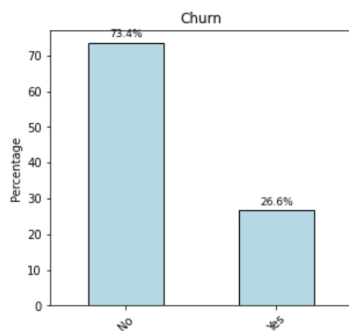
Machine learning approaches have become essential tools in predicting customer churn. Supervised machine learning techniques form the foundation of most churn prediction systems. Logistic regression, widely recognized for its computational simplicity and model interpretability, remains a popular choice for binary classification problems (Burez and Van den Poel). Decision trees and their ensemble variations, such as Random Forests, provide robust performance by capturing non-linear patterns and aggregating results across multiple classifiers (Lalwani et al.). In recent years, ensemble learning techniques have gained prominence. Boosting methods, such as AdaBoost and XGBoost, have been particularly effective in addressing class imbalance—a common issue in churn datasets. Studies highlight the superior predictive performance of XGBoost than traditional methods. Ensemble techniques not only improve prediction outcomes but also mitigate overfitting by aggregating predictions from multiple models (Lazarov and Capota; Lalwani et al.). Thus, logistic regression, decision tree, random forest, XGBoost and AdaBoost are potential methods that can be used in our project.

II. METHODOLOGY

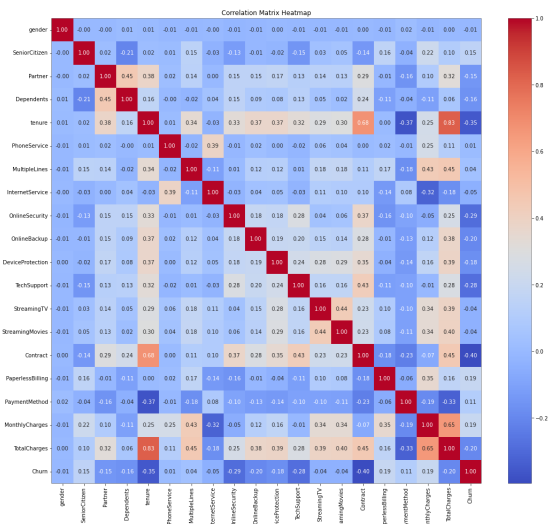
This project focuses on utilizing the Telco Customer Churn dataset (<https://huggingface.co/datasets/scikit-learn/churn-prediction>) from Hugging Face to explore churn analysis and prediction. This dataset includes the following information: CustomerID: A unique ID that identifies each customer. Demographic info about customers: Gender: The customer's gender: Male, Female. SeniorCitizen: Indicates if the customer is 65 or older: Yes, No. Partner: Indicates if the customer has any partners: Yes, No. Dependents: Indicates if the customer lives with any dependents: Yes, No. Dependents could be children, parents, grandparents, etc. Services that each customer has signed up for: PhoneService: Indicates if the customer subscribes to home phone service with the company: Yes, No. MultipleLines: Indicates if the customer subscribes to multiple telephone lines with the company: Yes, No. InternetService: Indicates if the customer subscribes to Internet service with the company: No, DSL, Fiber Optic, Cable. OnlineSecurity: Indicates if the customer subscribes to an additional online security service provided by the company: Yes, No. OnlineBackup: Indicates if the customer subscribes to an additional online backup service provided

by the company: Yes, No. DeviceProtection: Indicates if the customer subscribes to an additional device protection plan for their Internet equipment provided by the company: Yes, No. TechSupport: Indicates if the customer subscribes to an additional technical support plan from the company with reduced wait times: Yes, No. StreamingTV: Indicates if the customer uses their Internet service to stream television programming from a third-party provider: Yes, No. The company does not charge an additional fee for this service. StreamingMovies: Indicates if the customer uses their Internet service to stream movies from a third-party provider: Yes, No. The company does not charge an additional fee for this service. Customer account information: Tenure: Indicates the total amount of months that the customer has been with the company by the end of the quarter specified above. Contract: Indicates the customer's current contract type: Month-to-Month, One Year, Two Year. PaperlessBilling: Indicates if the customer has chosen paperless billing: Yes, No. PaymentMethod: Indicates how the customer pays their bill: Bank Withdrawal, Credit Card, Mailed Check. MonthlyCharges: Indicates the customer's current total monthly charge for all their services from the company. TotalCharges: Indicates the customer's total charges, calculated to the end of the quarter specified above. Churn: Indicates if the customer left within the last month: Yes, No.

In this dataset, the churn is the target variable, and other variables except CustomerID are input variables. This dataset contains both numerical and categorical variables. The dataset quality is generally good, with minimal preprocessing required. The variable TotalCharges was incorrectly identified as an object type due to non-numeric entries. After converting it to a numerical variable, we found 12 missing values, all corresponding to customers with tenure = 0. Since tenure = 0 only occurs when TotalCharges is NA in the whole dataset, this pattern indicates that the missing TotalCharges values are associated with customers who just started their service in less than a month, so there are no total charges yet. These missing values can be replaced with 0 or removed without impacting the analysis. Other variables show no missing values or data quality issues.



Additionally, there are potential challenges we need to deal with. Firstly, the dataset has significantly fewer churn cases (26.6 percent) compared to non-churn cases (73.4



Variable	VIF
const	35.13
gender	1.0
SeniorCitizen	1.15
Partner	1.46
Dependents	1.38
tenure	7.51
PhoneService	1.63
MultipleLines	1.4
InternetService	1.83
OnlineSecurity	1.28
OnlineBackup	1.22
DeviceProtection	1.3
TechSupport	1.33
StreamingTV	1.45
StreamingMovies	1.45
Contract	2.5
PaperlessBilling	1.21
PaymentMethod	1.19
MonthlyCharges	5.19
TotalCharges	10.72
Churn	1.38

percent), making churn prediction challenging. Imbalanced classes could lead models to consistently predict the majority class and have poor performance on the minority class. Secondly, there is a high correlation (0.83) between Tenure and TotalCharges and a high VIF (10.716) for TotalCharges, which suggests multicollinearity that needs to be dealt with. For other variables, there are no correlations greater than 0.8 or no VIF greater than 10.

For churn classification, logistic regression, decision tree, random forest, and XGBoost are methods to be used in this project. To prepare the dataset for the following modeling, categorical variables are label encoded from original categorical values to corresponding numerical values. Logistic regression is a statistical model used for binary classification that predicts the probability of an outcome, which is a simple and straightforward algorithm that can produce interpretable results. To prepare the dataset for logistic regression, numeric variables are scale transformed because variables have significantly different ranges. To handle multicollinearity, we can start with dropping variables with highest VIF and compare the performance before and after. To handle imbalance, we can use `class_weight` and oversampling. `class_weight` adjusts the penalty for misclassifying

classes within the loss function. Oversampling balances the dataset by generating synthetic samples for the minority class. The following three methods are robust to multicollinearity, so we do not need to preprocess it. Decision tree is a non-linear model that splits data into subsets based on feature thresholds, which handles non-linearity and interactions between features well and interpretable. To handle imbalance, we can use `class_weight` and oversampling. Class weights amplifies the influence of minority class samples in impurity calculations, ensuring the Decision Tree pays equal attention to both classes. Random Forest is an ensemble of decision trees that improves performance by averaging predictions and reducing overfitting. To handle imbalance, we can use `class_weight` and oversampling. XGBoost is a gradient boosting algorithm that sequentially builds trees to minimize error, incorporating improvements at each step. To handle imbalance, we can use the `scale_pos_weight` and oversampling. `scale_pos_weight` increases the contribution of the minority class in the loss function, forcing the model to pay more attention to correctly classifying positive samples.

III. RESULTS

To enhance the performance, we apply a systematic parameter optimization process known as Grid Search. This technique involves exploring a predefined search space of hyperparameters to identify the optimal combination for our specific classification problem. By conducting an exhaustive search over a range of possible parameter values, Grid Search facilitates the fine-tuning of the model, ultimately resulting in a more accurate and efficient classifier.

The model's performance will be assessed using various metrics, including precision, recall, f1-score and accuracy. Precision measures the proportion of correctly predicted positive observations out of all observations predicted as positive. Recall measures the proportion of correctly predicted positive observations out of all actual positives. F1-score combines precision and recall and is useful when the dataset is imbalanced. Accuracy measures how often the model correctly predicts the class label. Since in this customer churn dataset, the majority of customers are non-churns and churns are a minority, a model can achieve a high accuracy by predicting all customers as non-churns, we will prioritize recall of class 1 to ensure the model can effectively identify customers at risk of churning, minimizing false negatives where churners might be overlooked. Thus, to choose the best model, the primary metric is recall of class 1, the secondary metric is f1-score of class 1, and the third metric is accuracy.

We will show results of logistic regression, decision tree, random forest, and XGBoost. The results include precision of class 1, which is the percentage of predictions for class that are correct, recall of class 1, which is percentage of actual instances of class 1 that are correctly predicted, f1-score of class 1, which is a balance between precision and recall for class 1, and accuracy, which is the percentage of correctly classified instances (both true positives and true negatives) out of the total instances.

Here is the result from the logistic regression.

Logistic Regression	Precision (1)	Recall (1)	F1-Score (1)	Accuracy
Without TotalCharges - No Weight/OS	0.62	0.54	0.58	0.80
Without TotalCharges - Class Weight	0.50	0.82	0.62	0.74
Without TotalCharges - OS	0.51	0.79	0.62	0.75
Without TotalCharges - Weight + OS	0.51	0.79	0.62	0.75
With TotalCharges - No Weight/OS	0.62	0.53	0.58	0.80
With TotalCharges - Class Weight	0.51	0.82	0.62	0.74
With TotalCharges - OS	0.52	0.81	0.63	0.76
With TotalCharges - Weight + OS	0.52	0.81	0.63	0.76

Feature	Coefficient
MonthlyCharges	0.717175
TotalCharges	0.344856
PaperlessBilling	0.326818
SeniorCitizen	0.269772
InternetService	0.127386
MultipleLines	0.083449
StreamingMovies	0.057934
PaymentMethod	0.048446
StreamingTV	0.035542
gender	0.007748
Partner	-0.006663
DeviceProtection	-0.082662
OnlineBackup	-0.114639
Dependents	-0.122974
TechSupport	-0.223203
OnlineSecurity	-0.231232
PhoneService	-0.731028
Contract	-0.748391
tenure	-1.086028

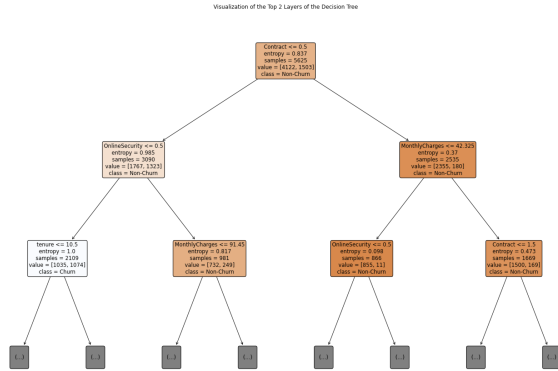
Model with TotalCharges and class weight is the best performance model among the 8 logistic regression models. Above are coefficients of the model. The logistic regression model highlights MonthlyCharges as the strongest positive predictor of churn (0.717) and TotalCharges as the second strongest positive predictor of churn (0.344), indicating that higher monthly costs significantly increase the likelihood of customer churn, likely due to price sensitivity. The PaperlessBilling coefficient (0.326818) suggests that customers opting for digital billing are more likely to churn, possibly due to its impersonal nature and unclear notifications, which can lead to dissatisfaction. The SeniorCitizen coefficient (0.269772) indicates that older customers are at a higher risk of churn, potentially due to affordability or technological barriers. In contrast, tenure has the most substantial negative effect (-1.086), showing that longer service durations are strongly associated with lower churn. Similarly, Contract (-0.748) reduces churn risk, as longer-term contracts provide higher stability and commitment and may include early termination fees. The coefficient for PhoneService (-0.731028) indicates a strong negative relationship with customer churn, which suggests that customers who have PhoneService are significantly less likely to churn compared to those without it. A potential reason for this could be that PhoneService provides utility and convenience. Additional features such as OnlineSecurity (-0.231) and TechSupport (-0.223) also show negative coefficients, suggesting that customers subscribing to these services are less likely to leave, reflecting their role in enhancing satisfaction and retention.

Here is the result from the decision tree.

Model with class weight is the best performance model among the 4 decision tree models. Above is the visualization

Decision Tree	Precision (1)	Recall (1)	F1-Score (1)	Accuracy
No Weight/OS	0.59	0.54	0.56	0.78
Class Weight	0.49	0.71	0.58	0.73
OS	0.52	0.61	0.56	0.75
Class Weight + OS	0.52	0.61	0.56	0.75

XGBoost	Precision (1)	Recall (1)	F1-Score (1)	Accuracy
No Weight/OS	0.65	0.49	0.56	0.80
Class Weight	0.52	0.79	0.63	0.75
OS	0.52	0.65	0.58	0.75
Class Weight + OS	0.52	0.65	0.58	0.75



of the top 2 layers of the model, revealing the most significant features for predicting customer churn. The root node starts with $\text{Contract} \leq 0.5$, which splits customers into two groups: those with month-to-month contracts and those with longer contracts. Customers with month-to-month contracts tend to churn more often. This group further splits based on $\text{OnlineSecurity} \leq 0.5$. Customers without OnlineSecurity are split again by $\text{tenure} \leq 10.5$, where shorter tenure customers have nearly equal churn and non-churn cases. For customers with OnlineSecurity, they are split further based on $\text{MonthlyCharges} \leq 91.45$. Customers with $\text{MonthlyCharges} \leq 91.45$ tend to not churn. Customers with longer contracts ($\text{Contract} > 0.5$) are split based on $\text{MonthlyCharges} \leq 42.325$. Those with charges below this tend to not churn, and their next split is $\text{OnlineSecurity} \leq 0.5$. Those without OnlineSecurity are prone to not churn. Those with charges > 42.325 are split by $\text{Contract} \leq 1.5$ into month-to-month and one-year contract and two-year contract groups, where customers with shorter contracts tend to be non-churn.

Here is the result from the random forest.

Random Forest	Precision (1)	Recall (1)	F1-Score (1)	Accuracy
No Weight/OS	0.65	0.48	0.55	0.80
Class Weight	0.55	0.74	0.63	0.77
OS	0.57	0.68	0.62	0.78
Class Weight + OS	0.54	0.73	0.62	0.77

Model with class weight is the best performance model among the 4 random forest models. Here are the seven most important features concluded by the model: 'Contract', 'tenure', 'TotalCharges', 'MonthlyCharges', 'TechSupport', 'OnlineSecurity', 'InternetService'. The model evaluates each feature's contribution to reducing impurity (Information Gain) across all decision trees, ranking features by their overall impact on predicting the target variable.

Here is the result from the XGBoost.

Based on the evaluation, the logistic regression with TotalCharges and with class weight appears to perform best in

customer churn prediction, which has precision of 0.51, recall of 0.82 (highest among all models), f1-score of 0.62, and accuracy of 0.74, making it highly effective at identifying churners while maintaining a good balance with precision and accuracy. The best parameters for this model is $C = 0.1$, $\text{penalty} = \text{liblinear}$.

IV. CONCLUSION

In this study, we analyzed customer churn in a telecom dataset using different machine learning models, including Logistic Regression, Decision Tree, Random Forest, and XGBoost. Addressing class imbalance played a critical role in improving model performance, where techniques like class weighting and oversampling ensured the minority class (churners) received enough attention. The results indicate that logistic regression with TotalCharges and class weighting has the best performance, with a precision (class 1) of 0.51, recall (class 1) of 0.82, f1-score (class 1) of 0.62, and accuracy of 0.74. This model was particularly effective at identifying customers at risk of churning while maintaining a good balance across the evaluation metrics. Combining insights from different models, key factors influencing customer churn are highlighted. High monthly charges, high total charges, paperless billing significantly increase the likelihood of churn, while longer tenure, longer contract type, phone service, online security, tech support enhance customer loyalty. These insights can help telecommunications companies develop more targeted customer retention strategies and have better customer management.

REFERENCES

- [1] J. Burez and D. Van den Poel, "Handling Class Imbalance in Customer Churn Prediction," *Expert Systems with Applications*, vol. 36, no. 3, 2009, pp. 4626–4636.
- [2] P. Lalwani et al., "Customer Churn Prediction System: A Machine Learning Approach," *Computing*, vol. 104, no. 2, 2022, pp. 271–294.
- [3] V. Lazarov and M. Capota, "Churn Prediction," *Technische Universität München*, 2021.
- [4] S. Tessitore, "What Is the Average Customer Churn Rate by Industry?," *CustomerGauge*, 2024.